



S.P.M College, Udantpuri

Bachelor Of Computer Application (BCA)

Part -2 (Paper-4)

– Hira Kumar

Computer Networking

RJ45 LAN Cable Wiring Color Code (T568A & T568B)

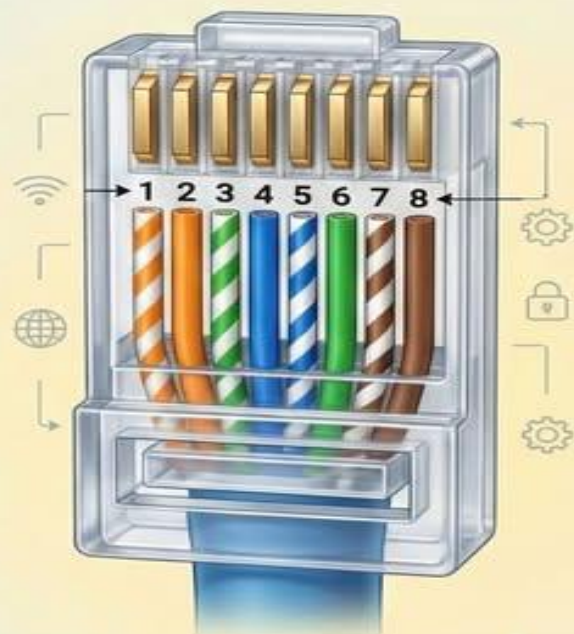
T568A Standard



T568A Color Code

Pin 1	White-Green
Pin 2	Green
Pin 3	White-Orange
Pin 4	Blue
Pin 5	White-Blue
Pin 6	Orange
Pin 7	White-Brown
Pin 8	Brown

T568B Standard



T568B Color Code

Pin 1	White-Orange
Pin 2	Orange
Pin 3	White-Green
Pin 4	Blue
Pin 5	White-Blue
Pin 6	Green
Pin 7	White-Brown
Pin 8	Brown

One Wrong Wire = No Internet

Case 1: Only Two Computers Connected via LAN/Ethernet Cable.



Connect Two Computers using LAN cable for Data Transfer and Communication



Step1 : Connect one end of the Ethernet/LAN cable to PC1.

Step2 : Connect the other end to PC2.

(Make sure both computers detect the ethernet connection.)

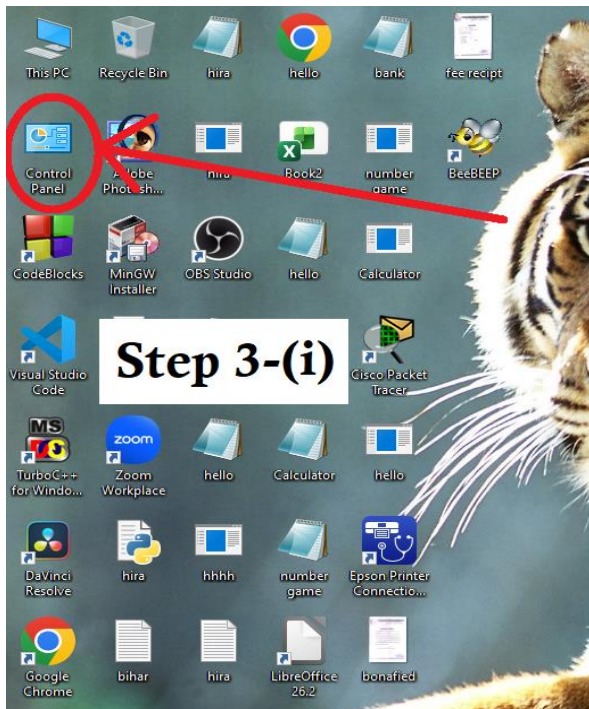


Step 3 : Configure PC1

- (i) Open **Control Panel**.
- (ii) Go to **Network and internet**.
- (iii) Go to **Network and sharing center**.
- (iv) Click **Change adapter settings**.
- (v) Right-click **Ethernet** → **Properties**.
- (vi) Select **Internet Protocol version 4 (TCP/IPv4)** → **Properties**.
- (vii) Select **Use the following IP address**.

IP Address : 192.168.1.1
Subnet mask : 255.255.255.0
Default Gateway : Leave blank
Preferred DNS : Leave blank

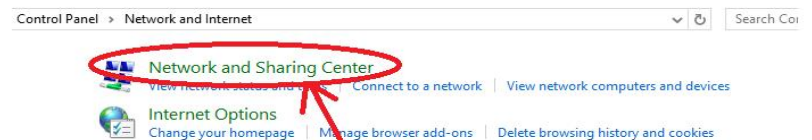
(viii) Click OK → Close



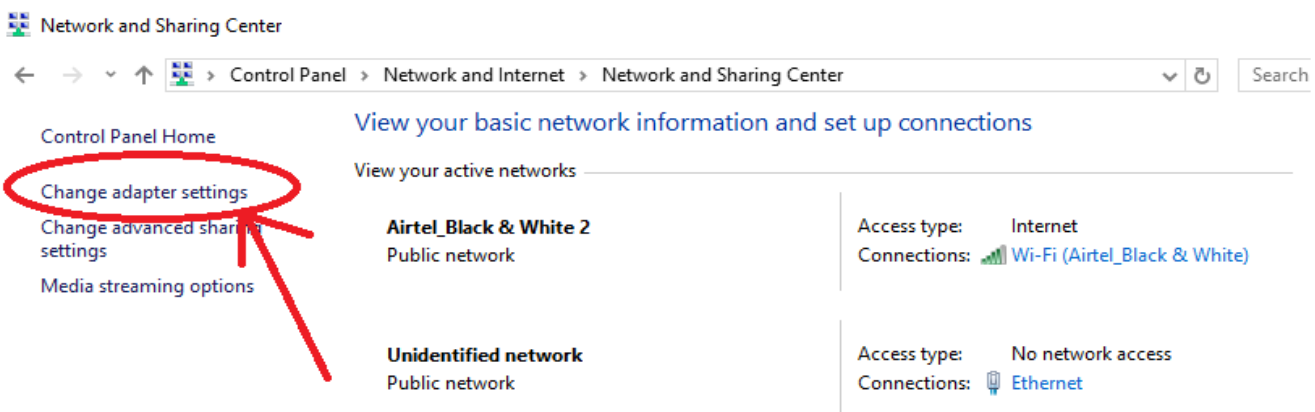
Step 3-(i)



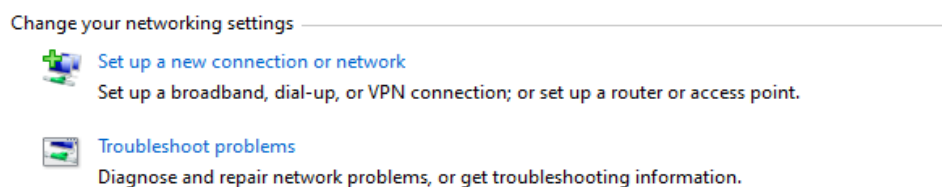
Step3 - (ii)



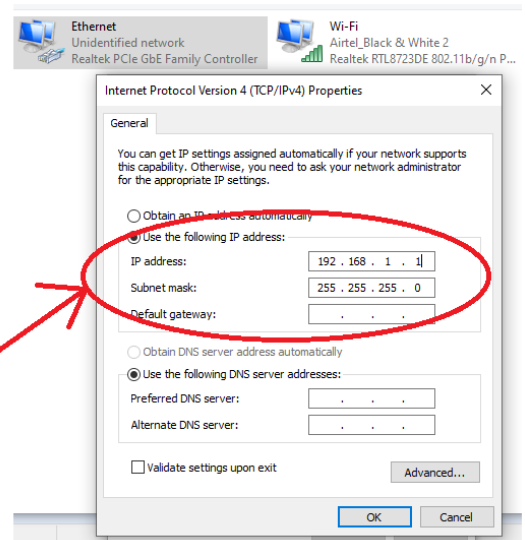
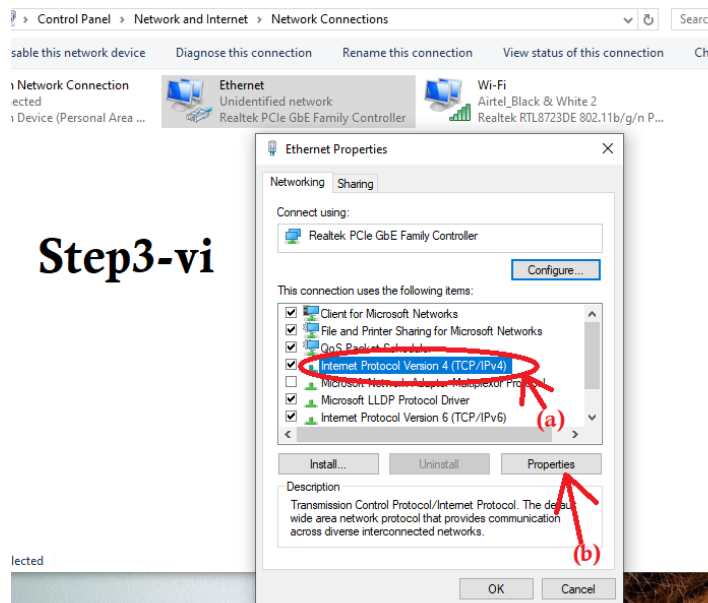
Step3-(iii)



Step3-iv



Step3 -v

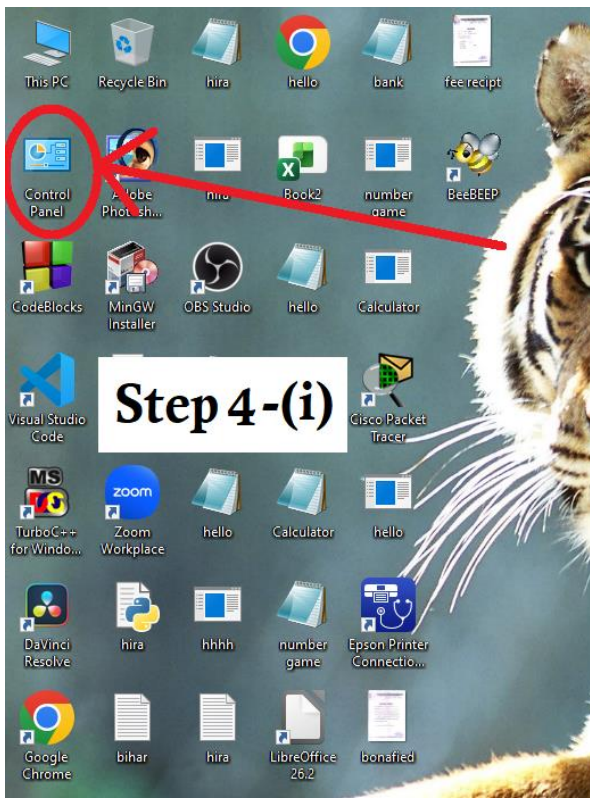


Step 4 : Configure PC2

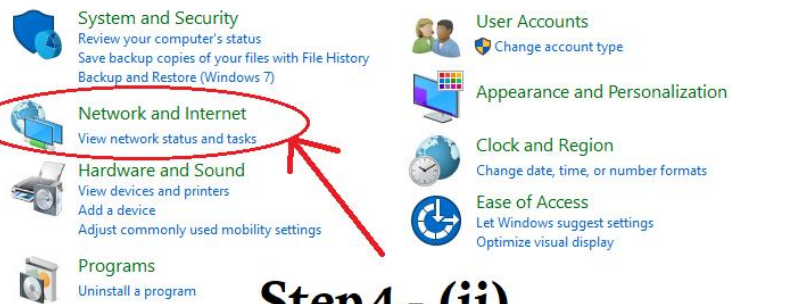
Follow the same steps on PC2.

- (i) Open **Control Panel**.
- (ii) Go to **Network and internet**.
- (iii) Go to **Network and sharing center**.
- (iv) Click **Change adapter settings**.
- (v) Right-click **Ethernet** → **Properties**.
- (vi) Select **Internet Protocol version 4 (TCP/IPv4)** → **Properties**.
- (vii) Select Use the following IP address.

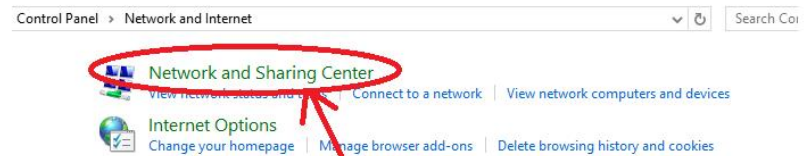
IP Address	:	192.168.1.2
Subnet mask	:	255.255.255.0
Default Gateway	:	Leave blank
Preferred DNS	:	Leave blank
- (viii) Click OK → Close



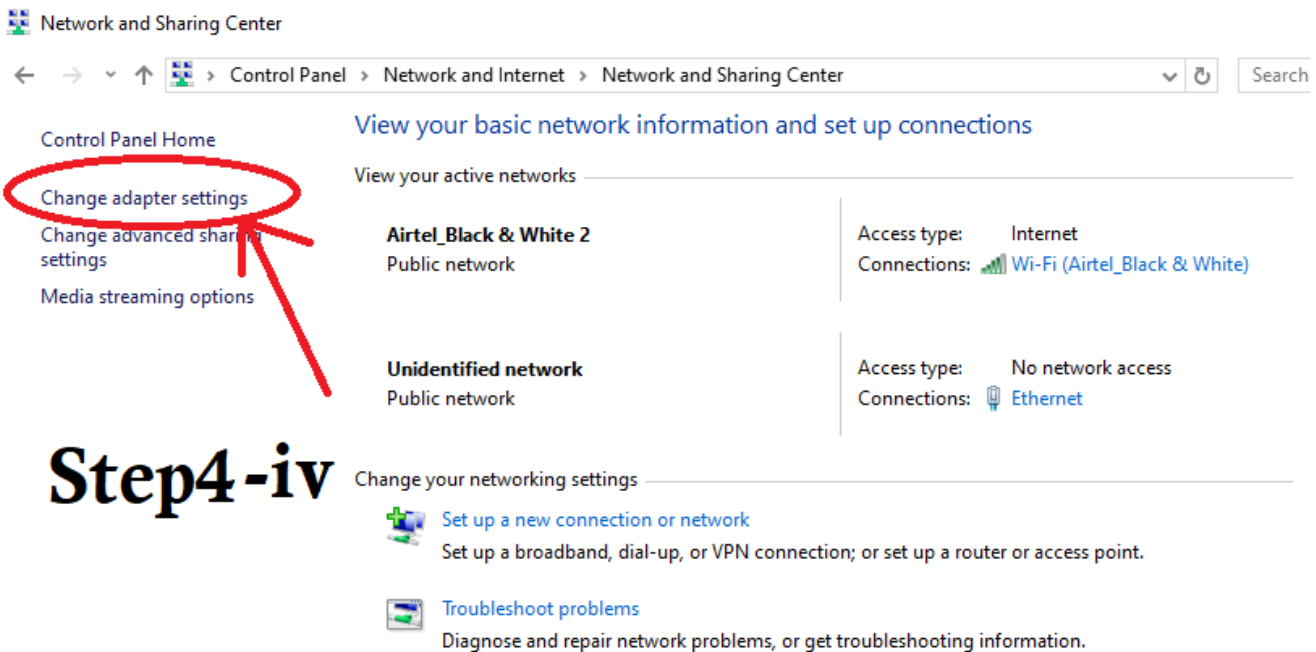
Step 4-(i)



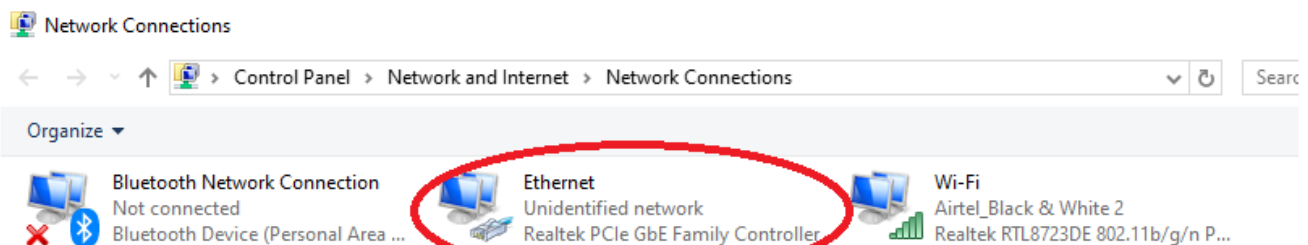
Step 4 - (ii)



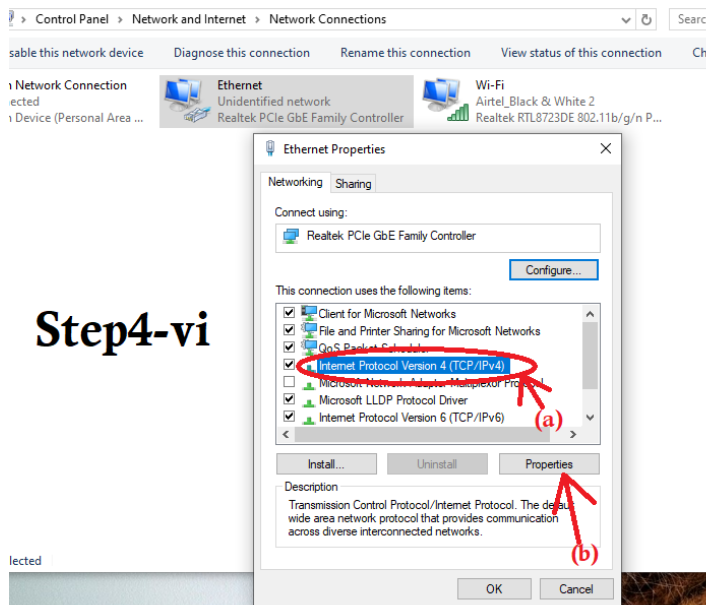
Step 4-(iii)



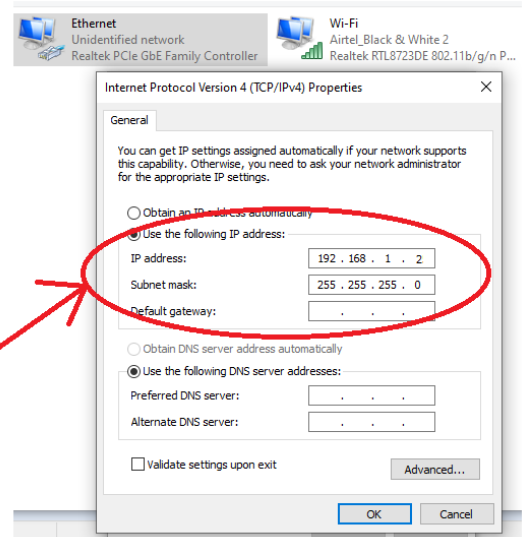
Step 4-iv



Step 4 - v



Step4-vi



Step4-vii

Note :

The IP addresses of the two computers must be different

PC1 : 192.168.1.1

PC2 : 192.168.1.2

But Subnet mask should be the same

255.255.255.0

Step 5 : Test the connection

On PC1, Open CMD (command prompt) and type :

Ping 192.168.1.2 (press enter)

```
C:\Users\abc>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Users\abc>
```



If you get this figure then PC1 and PC2 successfully communicating.

ON PC2, open CMD and type

Ping 192.168.1.1

```
C:\Users\hira>ping 192.168.1.1
```

```
Pinging 192.168.1.1 with 32 bytes of data:
```

```
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
```

```
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
```

```
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
```

```
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
```

```
Ping statistics for 192.168.1.1:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

```
Approximate round trip times in milli-seconds:
```

```
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

```
C:\Users\hira>
```



If you get this figure then PC2 and PC1 successfully communicating.

This Error message show if not successfully communicating

```
C:\Users\hira>ping 192.168.1.1
```

```
Pinging 192.168.1.1 with 32 bytes of data:
```

```
Request timed out.
```

```
Request timed out.
```

```
Reply from 192.168.1.2: Destination host unreachable.
```

```
Reply from 192.168.1.2: Destination host unreachable.
```

```
Ping statistics for 192.168.1.1:
```

```
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
```



Error Solve this way (see next page)

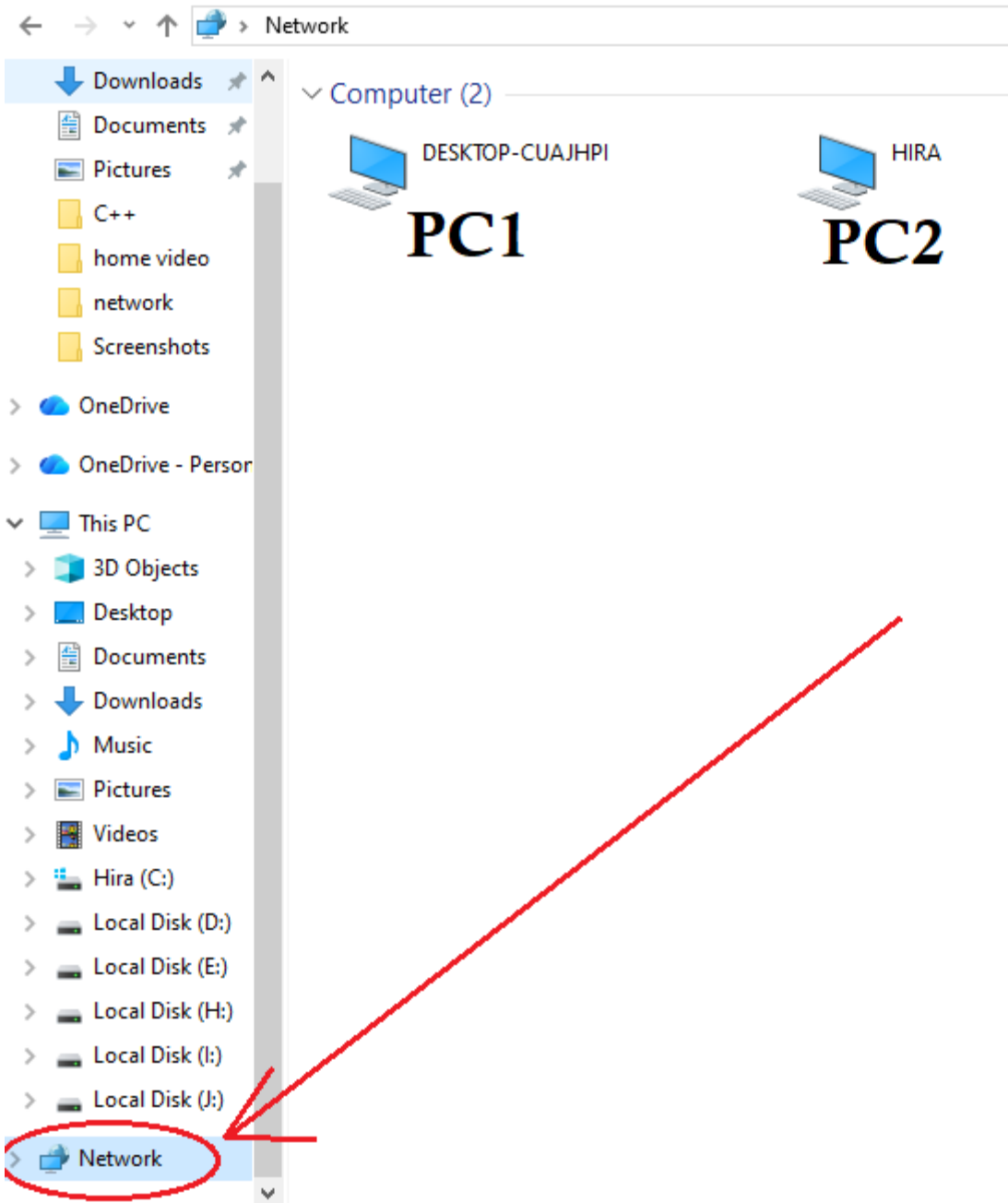
If Successfully connect then next step

Step 6 : Share files

If you want to share a folder from PC1 to PC2 or PC2 to PC1

- (i) Create a folder on PC
- (ii) Right – click the folder → properties.
- (iii) Go to sharing – Advance Sharing
- (iv) Check share this folder
- (v) Set the required permission.
- (vi) Click Apply → OK

Now, on PC1/PC2 open My computer and click network section.



Error Solve

Step1 : Open Network discovery and on file sharing

- (i) Open **Control Panel**.
- (ii) Go to **Network and internet**.
- (iii) Go to **Network and sharing center**.
- (iv) Click **Change advance sharing setting**.
- (v) Go to network discovery and on network discovery and file sharing

(vi) Go to All networks and turn on sharing and off password protected.

(vii) Click – Save change.

Network and Sharing Center

Control Panel Home

View your basic network information and set up connections

View your active networks

Network Name	Access type	Connections
Airtel_Black & White 2 Public network	Internet	Wi-Fi (Airtel_Black & White)
Unidentified network Public network	No Internet access	Ethernet

Change your networking settings

- [Set up a new connection or network](#)
Set up a broadband, dial-up, or VPN connection; or set up a router or access point.
- [Troubleshoot problems](#)
Diagnose and repair network problems, or get troubleshooting information.

1.

Change sharing options for different network profiles

Windows creates a separate network profile for each network you use. You can choose specific options for each profile.

Private

Guest or Public (current profile)

Network discovery

When network discovery is on, this computer can see other network computers and devices and is visible to other network computers.

☒ Turn on network discovery ✓
☐ Turn off network discovery

File and printer sharing

When file and printer sharing is on, files and printers that you have shared from this computer can be accessed by people on the network.

☒ Turn on file and printer sharing ✓
☐ Turn off file and printer sharing

All Networks

2.

Save changes Cancel

Check again CMD and ping command
If again error then –

Step 2 : Off Windows defender firewall

Network and Sharing Center

Control Panel > Network and Internet > Network and Sharing Center

Control Panel Home

- Change adapter settings
- Change advanced sharing settings
- Media streaming options

View your basic network information and set up connections

View your active networks

Airtel_Black & White 2 Public network	Access type: Internet Connections: Wi-Fi (Airtel_Black & White)
Unidentified network Public network	Access type: No network access Connections: Ethernet

Change your networking settings

- [Set up a new connection or network](#)
Set up a broadband, dial-up, or VPN connection; or set up a router or access point.
- [Troubleshoot problems](#)
Diagnose and repair network problems, or get troubleshooting information.

See also

- [Internet Options](#)
- [Windows Defender Firewall](#)

Windows Defender Firewall

Control Panel > System and Security > Windows Defender Firewall

Control Panel Home

- Allow an app or feature through Windows Defender Firewall
- [Change notification settings](#)
- [Turn Windows Defender Firewall on or off](#)
- [Restore defaults](#)
- [Advanced settings](#)
- Troubleshoot my network

Help protect your PC with Windows Defender Firewall

Windows Defender Firewall can help prevent hackers or malicious software from gaining access to your PC through the Internet or a network.

Update your Firewall settings

Windows Defender Firewall is not using the recommended settings to protect your computer.

[What are the recommended settings?](#)

[Use recommended settings](#)

Private networks	Not connected
Guest or public networks	Connected

Networks in public places such as airports or coffee shops

Windows Defender Firewall state:	Off
Incoming connections:	Block all connections to apps that are not on the list of allowed apps
Active public networks:	- Airtel_Black & White 2 - Unidentified network
Notification state:	Notify me when Windows Defender Firewall blocks a new app

See also

[Security and Maintenance](#)

Customize settings for each type of network

You can modify the firewall settings for each type of network that you use.

Private network settings



☐ Turn on Windows Defender Firewall

☐ Block all incoming connections, including those in the list of allowed apps

☒ Notify me when Windows Defender Firewall blocks a new app



☒ Turn off Windows Defender Firewall (not recommended)

Public network settings



☐ Turn on Windows Defender Firewall

☐ Block all incoming connections, including those in the list of allowed apps

☒ Notify me when Windows Defender Firewall blocks a new app



☒ Turn off Windows Defender Firewall (not recommended)

Check again CMD and ping command
If error again then

Step 3: run command prompt on windows power shell

Set-NetConnectionProfile -InterfaceAlias "Ethernet" -
NetworkCategory Private

```
Administrator: Windows PowerShell
PS C:\Users\hira> Set-NetConnectionProfile -InterfaceAlias "Ethernet" -NetworkCategory Private
PS C:\Users\hira>
```

Q. What is Lan Tester ?

A LAN Tester (also called a Network Cable Tester) is a device used to check whether an Ethernet/LAN cable is correctly wired and working properly.



Q. What is LAN Tester with Wire Tracker

A LAN Tester with Wire Tracker is a network cable testing device that does two main jobs:

1. LAN Cable Testing.
2. Wire Tracking (The wire tracker helps you find/identify a particular cable when many cables are present)



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